

Lab 5: Data from Multiple Tables (8% of total grade)

Submission: Use the included .sql file to put your answers in, then upload only this PDF file to Blackboard (Assessments > Lab 5 - Multiple Tables).

Name your file: **HTTP5126-L5-MultipleTables-LastNameFirstName.pdf**, replace *LastNameFirstName* with your name as displayed in Blackboard.

Purpose: To practice data retrieval from multiple tables by joining tables in result sets.

Requirements: For this assignment, you will use the **updated** Pet Store data tables.

NOTE: Run your queries on your database to make sure desired results are retrieved. Also import and execute your sql file to ensure it runs all your queries before submitting.

Pre-Lab:

1. Start your mySQL server and open phpMyAdmin.
2. Create a database for this lab (eg. http5126_lab5). Set the collation as utf8_unicode_ci.
3. Select the DB then import and execute the SQL file 'lab5_pet_store_tables.sql'. This should create 3 tables in your database (employee, stock_item, & sale) which will be used for the queries you create below.

Part 1: What are the sales for a particular item? (1%)

- A. Select all of the rows from the sale table with an stock item id value of 1014.

```
SELECT *  
FROM `sales`  
INNER JOIN stock_items ON sales.stock_item_id = stock_items.id  
WHERE stock_items.id = 1014;
```

- B. Select date (from sale), item name (from stock_item) to get the items sold with an id value of 1014.

```
SELECT sales.date, stock_items.name  
FROM sales  
INNER JOIN stock_items ON sales.stock_item_id = stock_items.id  
WHERE stock_items.id = 1014;
```

Part 2: What are the sales for a particular team member? (1%)

- A. Select all of the rows from the sale table with an employee id value of 111.

```
SELECT *  
FROM sales  
INNER JOIN employees ON sales.employee_id = employees.id  
WHERE employees.id = 111;
```

- B. Select date (from sale), first and last name (from employee), item id (from sale) to show the sales for the employee with an id of 111.

```
SELECT sales.`date`, employees.first_name, employees.last_name, sales.stock_item_id
FROM sales
INNER JOIN employees ON sales.employee_id = employees.id
WHERE employees.id = 111;
```

Part 3: Recent sales (1%)

- A. List the date (from sale) and item name (from stock_item) for the last 5 items sold. Ensure your results show the most recent sale first.

```
SELECT sales.date, stock_items.name
FROM sales
INNER JOIN stock_items ON sales.stock_item_id = stock_items.id
ORDER BY sales.date DESC
LIMIT 5;
```

- B. For the 5 most recent sales, display the first and last name of the employee (from employee) who processed them. Also include the date of the sale (from sale) ordered from most recent to least recent. And the name and price of the item (from stock_item).

```
SELECT sales.date, employees.first_name, employees.last_name,
stock_items.name, stock_items.price
FROM sales
INNER JOIN employees ON sales.employee_id = employees.id
INNER JOIN stock_items ON sales.stock_item_id = stock_items.id
ORDER BY sales.date DESC
LIMIT 5;
```

Part 4: What a week! and Go Team! (2%)

- A. Manager: "I need to review sales activity for a specific period. Can you provide me with a list of all sales made between 12 and January 18 (inclusive) of this year? The report should include the date of each sale, the item sold, and the first name of the employee who processed the transaction."

```
SELECT sales.date, stock_items.name, employees.first_name
FROM sales
```

```
INNER JOIN employees ON sales.employee_id = employees.id
INNER JOIN stock_items ON sales.stock_item_id = stock_items.id
WHERE sales.date BETWEEN '2025-01-12' AND '2025-01-18';
```

- B. Manager: "I want to see how well each salesperson is performing. Can you provide a report that shows the total number of sales made by each employee, grouped by their first and last name? The results should be ordered from the highest number of sales to the lowest."

```
SELECT employees.first_name, employees.last_name, COUNT(*) AS total_sales
FROM sales
INNER JOIN employees ON sales.employee_id = employees.id
GROUP BY employees.id
ORDER BY total_sales DESC;
```

Part 5: Challenge (2%)

- A. Manager: "I want to take a closer look at our top-performing salesperson. For all sales made by the top-seller, provide the date of the sale, the item name, its price, and category, along with the employee's first name." *Use aliases to rename your tables so your query is shorter to write. For this and all future labs you may use these table aliases in queries if you would like to.*

```
SELECT sa.date, items.name, items.price, items.category, emp.first_name
FROM sales sa
INNER JOIN employees emp ON sa.employee_id = emp.id
INNER JOIN stock_items items ON sa.stock_item_id = items.id
WHERE emp.id = 114
ORDER BY sa.date;
```

- B. Manager: "I need a report that shows the 'Times Sold' for each item. Please include the item ID, item name, price, and category for each item, even if it hasn't been sold. The results should be ordered by item ID."

```
SELECT stock_items.id, stock_items.`name`, stock_items.price,
stock_items.category, COUNT(sales.stock_item_id) AS times_sold
FROM stock_items
LEFT JOIN sales ON stock_items.id = sales.stock_item_id
GROUP BY stock_items.id
```

ORDER BY stock_items.id;

Part 6: Challenge Yourself (1%)

Create your own “human focused question” with the data set provided. Then write a query to answer the question. The question should be written as if you are either a customer of the store or an employee of the store (Manager, Sales Associate, Stock Person). Refer to the way some of the questions are written in this and earlier labs for an idea of how to write your “human focused question”.

Your query should use at least 1 JOIN and should be ordered so data is easy to parse. Preferably you should use an aggregate function to make your query more interesting.

- A. Write the question you have created next to the comment on line 35 of the provided sql file. eg. “-- A <your question here>”.

VP of Culture and Accountability: “Hi team!!! I need to pull a quick report showing total revenue generated per employee for last month, ordered lowest to highest. This is routine and is not related to the restructuring or the insurance claim. If your name is near the top, I wouldn’t worry about it. I would maybe update LinkedIn, but I wouldn’t worry about it. Thanks so much!”

- B. Create the query that returns the result your question asks for.

```
SELECT em.first_name, em.last_name, SUM(st.price) AS total_revenue
FROM sales sa
INNER JOIN employees em ON sa.employee_id = em.id
INNER JOIN stock_items st ON sa.stock_item_id = st.id
GROUP BY em.first_name, em.last_name
ORDER BY total_revenue ASC;
```